Lecture 14

Variation of Parameters

In some instances, we may want to solve a system of linear equations (such as when determining the coefficients of a particular solution). There is a technique that can help with this through the use of determinants of matrices called Cramer's Rule.

Suppose we want to solve a system of two linear equations in two unknowns

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1 \\ a_{21}x_1 + a_{22}x_2 = b_2 \end{cases}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} ae+bf \\ ce+df \end{bmatrix}$$

$2\times 2 \quad 2\times 1 \quad 2\times 1$

This can be rewritten as a matrix equation

term row colum

row

$$\overbrace{\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}}^{\text{colums}} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \leftarrow \text{constant}$$

$2\times 2 \quad 2\times 1 \quad 2\times 1$

unknown terms

As long as the coefficient matrix

coefficient

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\det\left(\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\right) = a_{11}a_{22} - a_{21}a_{12}$$

has nonzero determinant, then **Cramer's Rule** gives the solution to the system as

$$x_1 = \frac{\det A_1}{\det A}, \quad x_2 = \frac{\det A_2}{\det A}$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad A_1 = \begin{bmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{bmatrix}, \quad A_2 = \begin{bmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{bmatrix}$$

coefficient

replace first column by $\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$

replace second column by $\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$

Example: Solve

$$\begin{aligned} 2x + y &= 1 \\ 3x - 2y &= 5 \end{aligned} \Rightarrow \begin{bmatrix} 2 & 1 \\ 3 & -2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

using Cramer's Rule.

$$A = \begin{bmatrix} 2 & 1 \\ 3 & -2 \end{bmatrix}, \quad \begin{bmatrix} x \\ y \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 5 \end{bmatrix}, \quad A_1 = \cancel{\begin{bmatrix} 1 & 1 \\ 5 & -2 \end{bmatrix}}$$

$$A_2 = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$$

$$x = \frac{\det A_1}{\det A} = \frac{1(-2) - 5(1)}{2(-2) - 3(1)} = \frac{-2-5}{-4-3} = \frac{-7}{-7} = 1$$

$$y = \frac{\det A_2}{\det A} = \frac{2(5) - 3(1)}{2(-2) - 3(1)} = \frac{7}{-4-3} = \frac{7}{-7} = -1$$

so $x = 1, y = -1$

Cramer's Rule can be written for systems of more than two linear equations in more than two unknowns. We will explore this in a future lecture when we want to solve higher order equations.

Definition: If y_1 and y_2 are differentiable functions, the function

$$W[y_1, y_2](t) = \det\left(\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix}\right) = y_1 y_2' - y_1' y_2$$

is called the **Wronskian** of y_1 and y_2.

Note: We will adopt the notation

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc$$

Theorem (Test for Linear Independence): If y_1 and y_2 are solutions to a homogeneous linear equation and are differentiable on some interval I, and if

$a(t)y''+b(t)y'+c(t)y=0$

$$W[y_1, y_2](t) = 0$$

plug in a value

at any point t on I, then y_1 and y_2 are linearly dependent on I, otherwise they are linearly independent.

Example: Consider $y_1 = \cos(t)$, $y_2 = \sin(t)$ $\quad y''+y=0$ solve $\quad r=\pm i$

$$W[y_1,y_2](t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{vmatrix} = \cos(t)\cos(t) - (-\sin(t))(\sin(t))$$

$$= \cos^2(t) + \sin^2(t)$$

$$= 1 \neq 0$$

One of the drawbacks of the method of undetermined coefficients is that it depends heavily on what form the forcing function $f(t)$ takes. One take-away from this method we can expand upon is the idea that a particular solution may be of a similar form to that of the homogeneous solution.

To that end, consider the nonhomogeneous linear 2nd-order differential equation

$$a(t)y'' + b(t)y' + c(t)y = f(t) \tag{14.1}$$

and let $y_1(t)$ and $y_2(t)$ be two linearly independent solutions to the corresponding homogeneous equation. Then the solution to the homogeneous equation can be written

$$y_h = c_1 y_1 + c_2 y_2$$

how can we find the particular solution?

As we have seen, sometimes the form of the particular solution is a product of a function with the solutions to the homogeneous equation. We expand this idea by seeking the particular solution to (14.1) of the form

$$y_p = v_1(t)y_1(t) + v_2(t)y_2(t)$$

where $v_1(t)$ and $v_2(t)$ are differentiable functions. If we want this to be a solution we must determine $v_1(t)$ and $v_2(t)$ so that y_p satisfies (14.1). We can do so by differentiating and substituting.

$= 0$ Goal: find v_1, v_2

$$y_p' = (v_1'y_1 + v_1y_1') + (v_2'y_2 + v_2y_2')$$

Method: use Cramer's Rule construct a system.

We make the assumption that $v_1'y_1 + v_2'y_2 = 0$ (Why?) and so

$$y_p' = v_1 y_1' + v_2 y_2'$$

and

$$y_p'' = (v_1'y_1' + v_1y_1'') + (v_2'y_2' + v_2y_2'')$$

Substituting into (14.1) we see that $a(t)y'' + b(t)y' + c(t)y = f(t)$

$$a(t)\underbrace{[(v_1'y_1' + v_1y_1'') + (v_2'y_2' + v_2y_2'')]}_{y_p''} + b(t)\underbrace{[v_1y_1' + v_2y_2']}_{y_p'} + c(t)\underbrace{[v_1y_1 + v_2y_2]}_{y_p}$$

$$\begin{aligned}
&= [a(t)v_1y_1'' + b(t)v_1y_1' + c(t)v_1y_1] \to 0 \\
&\quad + [a(t)v_2y_2'' + b(t)v_2y_2' + c(t)v_2y_2] \to 0 \\
&\quad + a(t)[v_1'y_1' + v_2'y_2']
\end{aligned}$$

$$= 0 + 0 + a(t)[v_1'y_1' + v_2'y_2'] \overset{(14.1)}{=} f(t)$$

Therefore if we divide by $a(t)$ we have

$$v_1'y_1' + v_2'y_2' = \frac{f(t)}{a(t)} \qquad (2)$$

This gives us two linear equations in the unknowns v_1' and v_2' as

matrix for Wronskian

$$\begin{cases} y_1 v_1' + y_2 v_2' = 0 \\ y_1' v_1' + y_2' v_2' = \dfrac{f(t)}{a(t)} \end{cases} \Rightarrow \begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix} \cdot \begin{bmatrix} v_1' \\ v_2' \end{bmatrix} = \begin{bmatrix} 0 \\ f(t)/a(t) \end{bmatrix}$$

We can solve this using Cramer's Rule. Cramer's Rule says that

$$v_1' = \frac{W_1}{W} \text{ and } v_2' = \frac{W_2}{W}$$

where

Wronskian

linearly independent

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \neq 0, \quad W_1 = \begin{vmatrix} 0 & y_2 \\ f(t)/a(t) & y_2' \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(t)/a(t) \end{vmatrix}$$

We can integrate to find v_1 and v_2.

no constants of integration!

$$v_1 = \int \frac{W_1}{W}\,dt \quad , \quad v_2 = \int \frac{W_2}{W}\,dt$$

This technique to find v_1 and v_2 so that $v_1 y_1 + v_2 y_2$ is a particular solution to (14.1) is called **Variation of Parameters**.

Variation of Parameters Solution Method: Consider the equation

$$a(t)y'' + b(t)y' + c(t)y = f(t)$$

1. Find two linearly independent solutions y_1 and y_2 to the associated homogeneous equation. So $y_h = c_1 y_1 + c_2 y_2$.
2. Compute the Wronskian W and W_1, and W_2

$\begin{bmatrix} 0 \\ f/a \end{bmatrix}$

3. Compute v_1 and v_2
4. A particular solution is $y_p = v_1 y_1 + v_2 y_2$

Once we have both the homogeneous solution and particular solution we have that the general solution is

$$y_g = y_h + y_p = (c_1 y_1 + c_2 y_2) + (v_1 y_1 + v_2 y_2)$$

Example: Find the general solution to

$f(t) = e^t \ln(t)$

$$y'' - 2y' + y = e^t \underline{\ln(t)}, \quad t > 0$$

$a = 1$

cannot use undetermined coefficients.

1. char eqn: $r^2 - 2r + 1 = 0$
$(r-1)^2 = 0 \Rightarrow r = 1, 1$

$y_h = c_1 \underline{e^t} + c_2 \underline{te^t}, \quad y_1 = e^t, y_2 = te^t$

2. Find W, W_1, W_2

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^t & te^t \\ e^t & e^t + te^t \end{vmatrix} = (e^{2t} + te^{2t}) - te^{2t} = e^{2t}$$

$$W_1 = \begin{vmatrix} 0 & y_2 \\ f/a & y_2' \end{vmatrix} = \begin{vmatrix} 0 & te^t \\ e^t \ln(t) & e^t + te^t \end{vmatrix} = 0 - te^{2t}\ln(t) = -te^{2t}\ln(t)$$

$$W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f/a \end{vmatrix} = \begin{vmatrix} e^t & 0 \\ e^t & e^t \ln(t) \end{vmatrix} = e^{2t}\ln(t) - 0$$

IBP
$\ln(t) \xrightarrow{+} -t$
$\frac{1}{t} \Rightarrow -\frac{1}{2}t^2$

3. $$v_1 = \int \frac{W_1}{W}\,dt = \int \frac{-te^{2t}\ln(t)}{e^{2t}}\,dt = \int -t\ln(t)\,dt = -\frac{1}{2}\ln(t)t^2 + \frac{1}{4}t^2$$

IBP
$\ln(t) \xrightarrow{+} 1$
$\frac{1}{t} \Rightarrow t$

$$v_2 = \int \frac{W_2}{W}\,dt = \int \frac{e^{2t}\ln(t)}{e^{2t}}\,dt = \int \ln(t)\,dt = t\ln(t) - t$$

So $y_p = v_1 y_1 + v_2 y_2 = (-\frac{1}{2}t^2\ln(t) + \frac{1}{4}t^2)e^t + (t\ln(t) - t)\,te^t$

Thus $\boxed{y_g = y_h + y_p = (c_1 e^t + c_2 te^t) + (\frac{1}{2}t^2 e^t \ln(t) - \frac{3}{4}t^2 e^t)}$

Example: Find the general solution to

$a=4$ $\qquad$ $f = \csc(3t)$

$$4y'' + 36y = \csc(3t)$$

1. char eqn: $4r^2 + 36 = 0$

$\Rightarrow r^2 = 9 \Rightarrow r = \pm 3i$

$y_h = c_1 \cos(3t) + c_2 \sin(3t)$

$y_1 = \cos(3t)$, $y_2 = \sin(3t)$

2. $W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos(3t) & \sin(3t) \\ -3\sin(3t) & 3\cos(3t) \end{vmatrix} = 3\cos^2(3t) + 3\sin^2(3t)$

$= 3$

$W_1 = \begin{vmatrix} 0 & y_2 \\ f/a & y_2' \end{vmatrix} = \begin{vmatrix} 0 & \sin(3t) \\ \frac{1}{4}\csc(3t) & 3\cos(3t) \end{vmatrix} = 0 - \frac{1}{4}\csc(3t)\sin(3t)$

$= -\frac{1}{4}$

$W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f/a \end{vmatrix} = \begin{vmatrix} \cos(3t) & 0 \\ -3\sin(3t) & \frac{1}{4}\csc(3t) \end{vmatrix} = \frac{1}{4}\csc(3t)\cos(3t) - 0$

$= \frac{1}{4}\cot(3t)$

3. $v_1 = \int \frac{W_1}{W} dt = \int \frac{-1/4}{3} dt = \int -\frac{1}{12} dt = -\frac{1}{12}t$

$v_2 = \int \frac{W_2}{W} dt = \int \frac{\frac{1}{4}\cot(3t)}{3} dt = \int \frac{1}{12}\cot(3t) dt$

$$= \int \frac{1}{12} \cdot \frac{\cos(3t)}{\sin(3t)}\,dt = \frac{1}{12} \cdot \frac{1}{3} \ln|\sin(3t)|$$

$$= \frac{1}{36} \ln|\sin(3t)|$$

$u = \sin(3t)$

$du = 3\cos(3t)\,dt$

4. So $y_p = v_1 y_1 + v_2 y_2$

$$= \left(-\frac{1}{12} t\right)(\cos(3t)) + \left(\frac{1}{36} \ln|\sin(3t)|\right)\sin(3t)$$

Thus

$$\boxed{y_g = y_h + y_p = (c_1 \cos(3t) + c_2 \sin(3t)) + \left(\frac{1}{36} \sin(3t) \ln|\sin(3t)| - \frac{1}{12} t \cos(3t)\right)}$$

Example: Find the general solution to

$f(t) = t^{-3}e^{3t}$

$$y'' - 6y' + 9y = t^{-3}e^{3t}, \quad t \neq 0$$

$a = 1$

1. char eqn: $r^2 - 6r + 9 = 0$

$\Rightarrow (r-3)^2 = 0, \quad r = 3, 3$

$y_h = c_1 e^{3t} + c_2 t e^{3t}$

So $y_1 = e^{3t}, \; y_2 = te^{3t}$

2. $W = \begin{vmatrix} e^{3t} & te^{3t} \\ 3e^{3t} & e^{3t} + 3te^{3t} \end{vmatrix} = (e^{6t} + 3te^{6t}) - (3te^{6t}) = e^{6t} \neq 0$

$W_1 = \begin{vmatrix} 0 & te^{3t} \\ t^{-3}e^{3t} & e^{3t} + 3te^{3t} \end{vmatrix} = 0 - (t^{-3}e^{3t})(te^{3t}) = -t^{-2}e^{6t}$

$W_2 = \begin{vmatrix} e^{3t} & 0 \\ 3e^{3t} & t^{-3}e^{3t} \end{vmatrix} = t^{-3}e^{6t} - 0 = t^{-3}e^{6t}$

3. $v_1 = \int \frac{W_1}{W}\,dt = \int \frac{-t^{-2}e^{6t}}{e^{6t}}\,dt = \int -t^{-2}\,dt = t^{-1}$

$v_2 = \int \frac{W_2}{W}\,dt = \int \frac{t^{-3}e^{6t}}{e^{6t}}\,dt = \int t^{-3}\,dt = -\frac{1}{2}t^{-2}$

4. $y_p = v_1 y_1 + v_2 y_2 = (t^{-1})e^{3t} + (-\frac{1}{2}t^{-2})(te^{3t}) = \frac{1}{2}t^{-1}e^{3t}$

Thus $\boxed{y_g = y_h + y_p = (c_1 e^{3t} + c_2 te^{3t}) + \frac{1}{2}t^{-1}e^{3t}}$